HKU MAIB7002 / GROUP Q

FRESH FOOD. SMARTER DISCOUNTS.



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1 Problem

Should we discount it tomorrow?
If so, by how much?

Fresh-food managers balance unsold perishables against value lost to markdowns. We forecast next-day sales, then compare supported prices to recommend a daily discount.

Too little discount
Unsold food

Too much discount
Value given away

Goal: recommend a specific, historically supported discount.

Data Cleaning

FreshRetailNet-50K provides daily sales, discounts and stockouts. Select regular sellers with repeated price changes.

15-85% days at rate <0.95; 6+ state switches; <20% zero-sales days.

50,000 histories → **16,684** eligible → **312** histories / 5 stores

Take the five stores with the most eligible histories.

- **Check daily records**
  Unique keys; no missing fields or date gaps.
- **Handle invalid price rates**
  Cap rates above 1; mask zero-price days.
- **Build past-only features**
  Exclude rows without complete history inputs.

25,896 train / validation rows + 2,184 final-week rows

2 Models

Five approaches, from a simple reference to a combined forecast.

10 INPUTS
Sales history, stockouts, discounts, activity and calendar → **TARGET** **Next-day normalized sales**

Tomorrow's candidate price rate is an input; the model does not directly predict the discount.

01 7-day mean

Use the previous week's average sales as tomorrow's forecast. This sets a simple benchmark.



Baseline | no fitted model

02 kNN

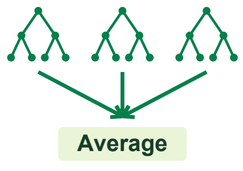
Find similar historical days and average their sales. Feature scaling makes distances comparable.



25 neighbours | standardized inputs | equal weights

03 Random Forest

Fit many trees on resampled data, then average their predictions to reduce variability.



300 trees | depth 10 | minimum leaf size 3

04 CatBoost

Add trees sequentially. Each new tree targets prediction errors left by the earlier trees.



500 trees | depth 5 | learning rate 0.03

05 RF + CatBoost

Average two strong forecasts. Equal weights keep the model simple; the validation curve is almost flat near 50/50.

0.5 RF + **0.5 CatBoost** → **Sales** forecast

EVALUATION Five expanding validation weeks choose settings → Final week: 26 Jun - 2 Jul 2024

3 Results & Implications

FORECAST RESULTS

	MAE	RMSE
7-day mean	0.3142	0.5209
kNN	0.3040	0.4900
Random Forest	0.2923	0.4646
CatBoost	0.2896	0.4603
RF + CatBoost	**0.2892**	**0.4593**

MAE = mean absolute error; both metrics: lower is better.

7.9% lower MAE

vs. the baseline; CatBoost alone is nearly tied.

FROM FORECAST TO DECISION

98.3% → **55.0%**

of cases recommend a discount

Model + support → + stock rules

Compare supported rates using predicted sales x price rate (zero cost). Stockout rules constrain the choice. One eligible product-day selects **5% off**.

Example: Store 18 / Product 11 / 26 June 2024

Recommendations remain untested in real stores.

IMPLICATION Use forecasts as decision support. Validate pricing effects in a controlled store pilot, with inventory and cost data.

Limits: five stores; retrospective selection; reused benchmark. Value is a proxy, not currency. No causal sales, revenue or waste gain shown.

Source: Dingdong FreshRetailNet-50K, CC BY 4.0. Final notebook: pricing-ml-ap @ a7e26dd. Decorative produce artwork generated with AI.